

What we're doing

Does MarFold predict contacts for a circularly permuted protein as well as it does for the natural, non-permuted parent — or does its accuracy come from having memorised the parent's sequence→contact mapping at fixed sequence separations?

Why

A circular permutation is close to the ideal probe for this. `1UN2` (CPDsbA-Q100T99, 2.4 Å) has the same fold, the same residues and essentially the same 3D contact set as wild-type *E. coli* DsbA, but the contact map in *sequence* coordinates is completely reorganised: the chain is cut between T99 and Q100 and the two halves are swapped, so every contact that spans the old N/C-termini changes sequence separation, and the $|i-j|$ distribution the model has learned as a prior no longer matches the structure.

Two outcomes are informative:

- CP $\approx$ WT. The model is reasoning about the fold from local sequence, and transfers it across a re-ordering it has never seen. Strong evidence against the memorisation reading that [#94](<https://github.com/Open-Athena/MarinFold/issues/94>) and [#213](<https://github.com/Open-Athena/MarinFold/issues/213>) probed from other directions. - CP $\ll$ WT. The model leans on a learned separation prior / on retrieving the parent, and a re-ordering breaks it. That would be a concrete, mechanistic instance of the generalisation gap.

Wild-type DsbA (UniProt `P0AEG4`) is in AFDB and therefore almost certainly in exp199's 70.9M-protein training set. The permutant is an engineered construct from 1999 and is not — but it is ~96% local-sequence-identical to the parent, so a pure nearest-neighbour predictor would still retrieve the right *local* contacts. The discriminating signal is the junction-spanning contacts: pairs that are in contact in 3D but whose sequence separation the permutation changed.

Results

The premise holds. CP and WT are the same fold: 88.5% of WT's contacts are present in CP (Jaccard 0.734), against a crystal-to-crystal floor of 0.85–0.88 from two WT replicates. 47% of WT's contacts cross the cut, and for those the separation transforms exactly as $\text{CP_sep} = 194 - \text{WT_sep}$.

The permutation costs ~ 0.09 R-precision; the extra residues cost nothing. Mean $\pm$ sd over 10 rollout seeds (exp82 recipe, exp89 metrics):

CP 1UN2, permuted — R-precision 0.5247 ± 0.0091 , AUC 0.952, 140.7 contacts/rollout

Control (WT + linker + tail, native order) — R-precision 0.6177 ± 0.0104 , AUC 0.983, 181.5 contacts/rollout

WT 1FVK / 1DSB / 1A2J — R-precision 0.626 / 0.617 / 0.631, AUC ~ 0.982 , ~ 179 contacts/rollout

The control has the permutant's exact length and non-native residues in wild-type order and scores like the real WT crystals, so the -0.0930 gap (Welch $t = -21.3$, $p = 4.9\text{e-}14$) is the re-ordering itself.

The damage lands exactly where the theory says

Same model, same pairs, same universe, split by whether the permutation changed that pair's sequence separation:

Separation UNCHANGED (within-segment) — CP — WT: R-precision -0.0506 , AUC -0.0020

Separation CHANGED (cross-segment) — CP — WT: R-precision -0.1205 , AUC -0.0448

Moved pairs lose 2.4x more R-precision and 22x more AUC than unmoved ones.

Conclusion

MarinFold is not merely retrieving the parent — but a circular permutation costs it a lot, and the cost is specific to re-ordered pairs.

Keeping R-precision 0.525 (vs 0.626 WT) and AUC 0.952 rules out the strong memorisation story: a lookup keyed on the parent at fixed separations would collapse, not degrade by 16%. But the gap is real, caused by the re-ordering, concentrated on moved pairs, and accompanied by a 22% drop in contacts emitted per rollout (no truncation — the model is declining to commit, not running out of budget).

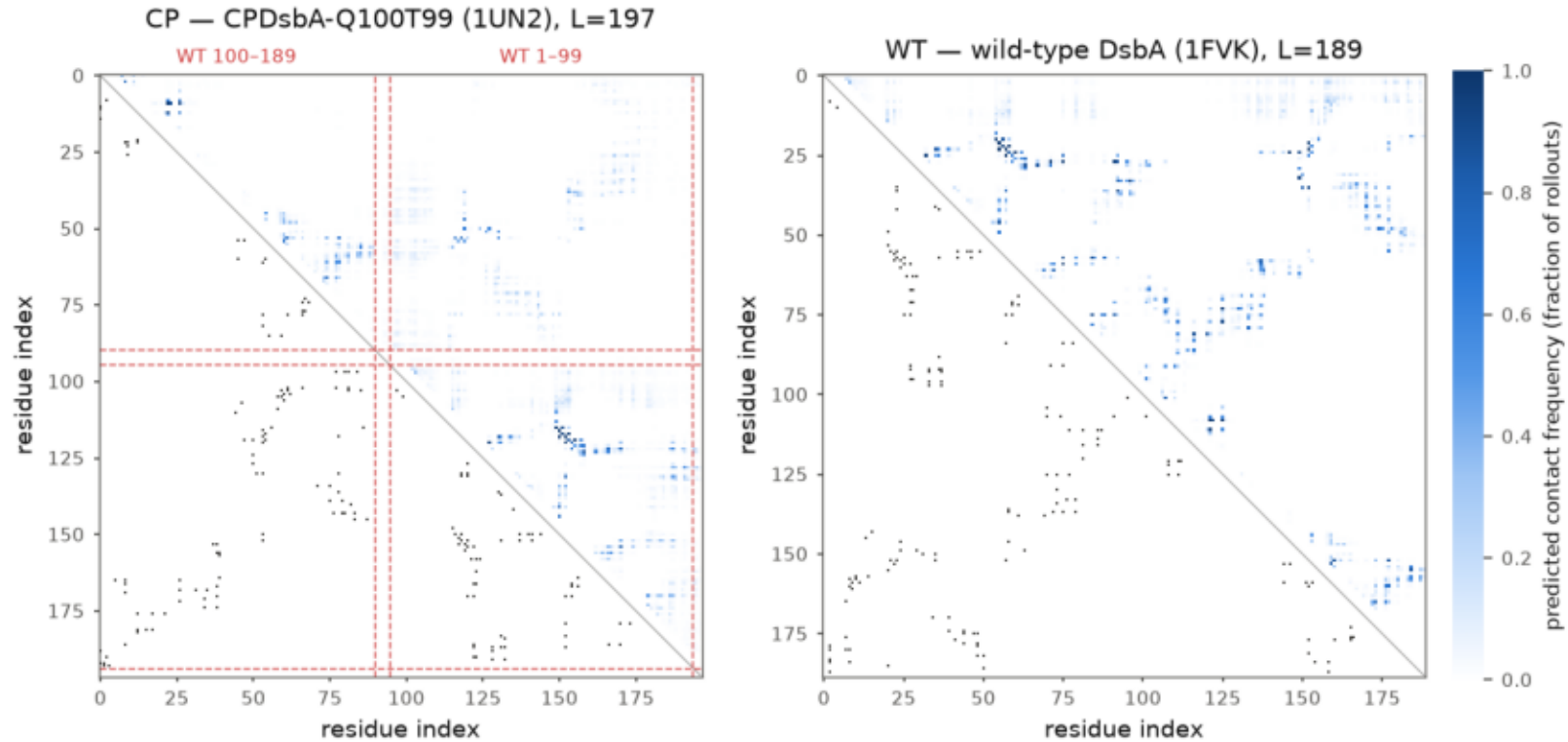
The picture is a model that has learned the fold *and* a strong prior over where in sequence its contacts should live, leaning on the prior more than a genuine folding model would. Follow-up: does permutation augmentation at training time close it, and does that transfer to ordinary proteins? Cheap to test — it needs no new structures.

Caveat: $n = 1$ permutant. The internal moved-vs-unmoved contrast carries the argument; the absolute gap rests on one pair of crystals.

contact_maps.png

Prediction (upper triangle) vs pyconfind truth (lower), one shared colour scale. The permutant's map is fainter because the model is less confident.

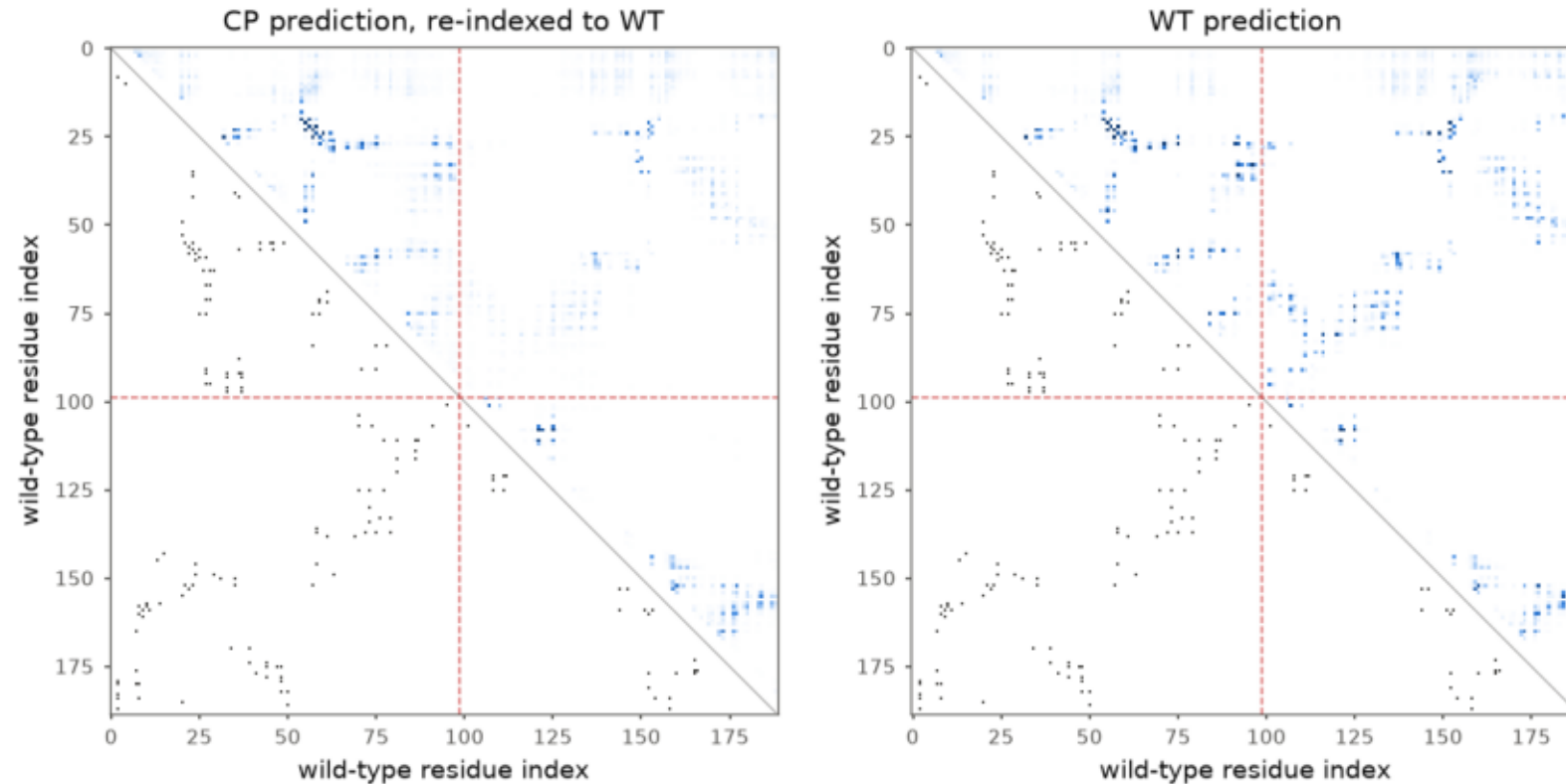
MarinFold contact prediction (upper) vs pyconfind ground truth (lower)



cp_in_wt_frame.png

Both predictions re-indexed into wild-type coordinates. The block straddling the T99/Q100 cut is visibly thinner in the permutant.

Both predictions in wild-type coordinates (lower triangle: WT truth)

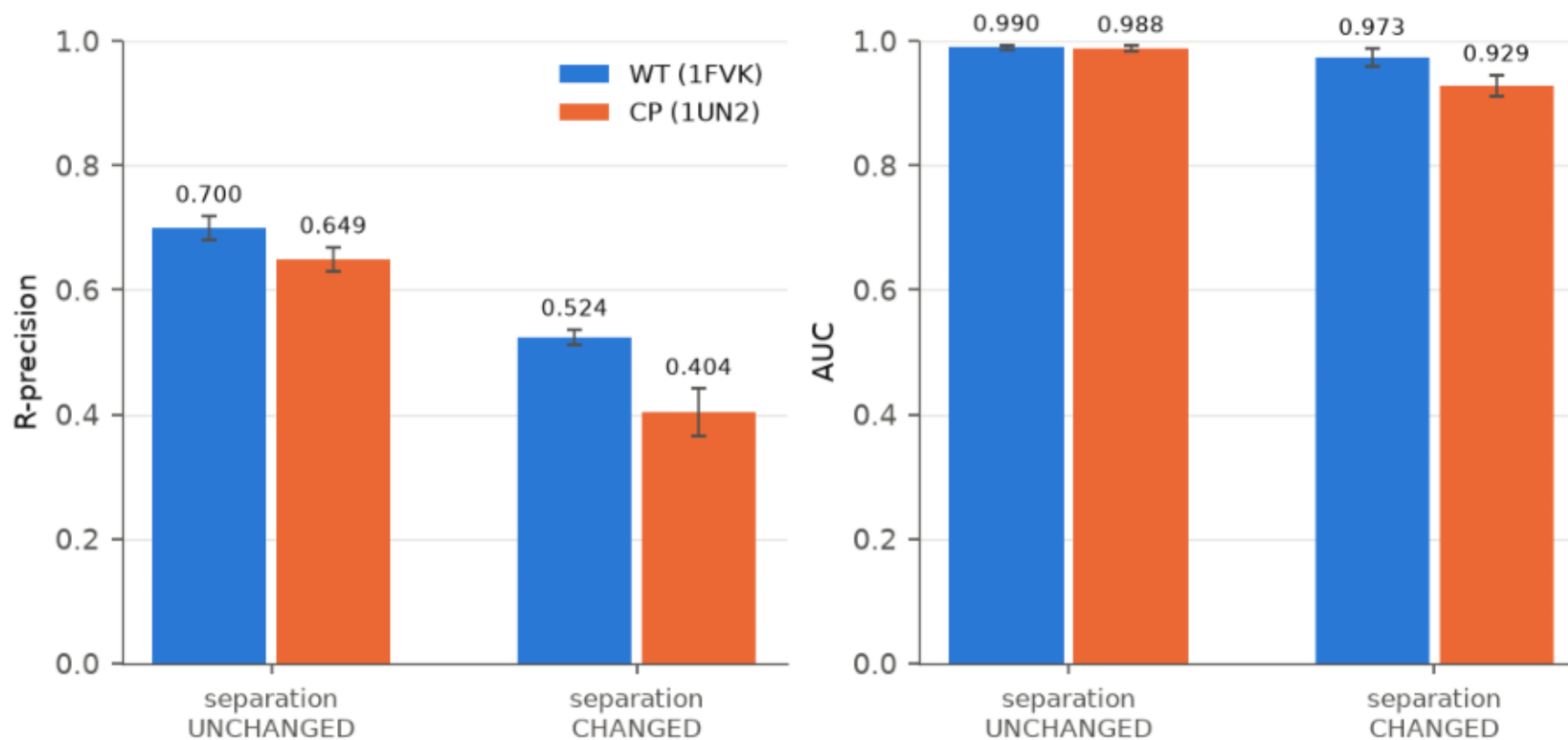


Red dashes mark the T99/Q100 cut. Off-diagonal blocks straddling it are the contacts the permutation moved to a different sequence separation.

permutation_contrast.png

Same pairs and same model, split by whether the permutation moved them. Moved pairs lose 2.4x more R-precision.

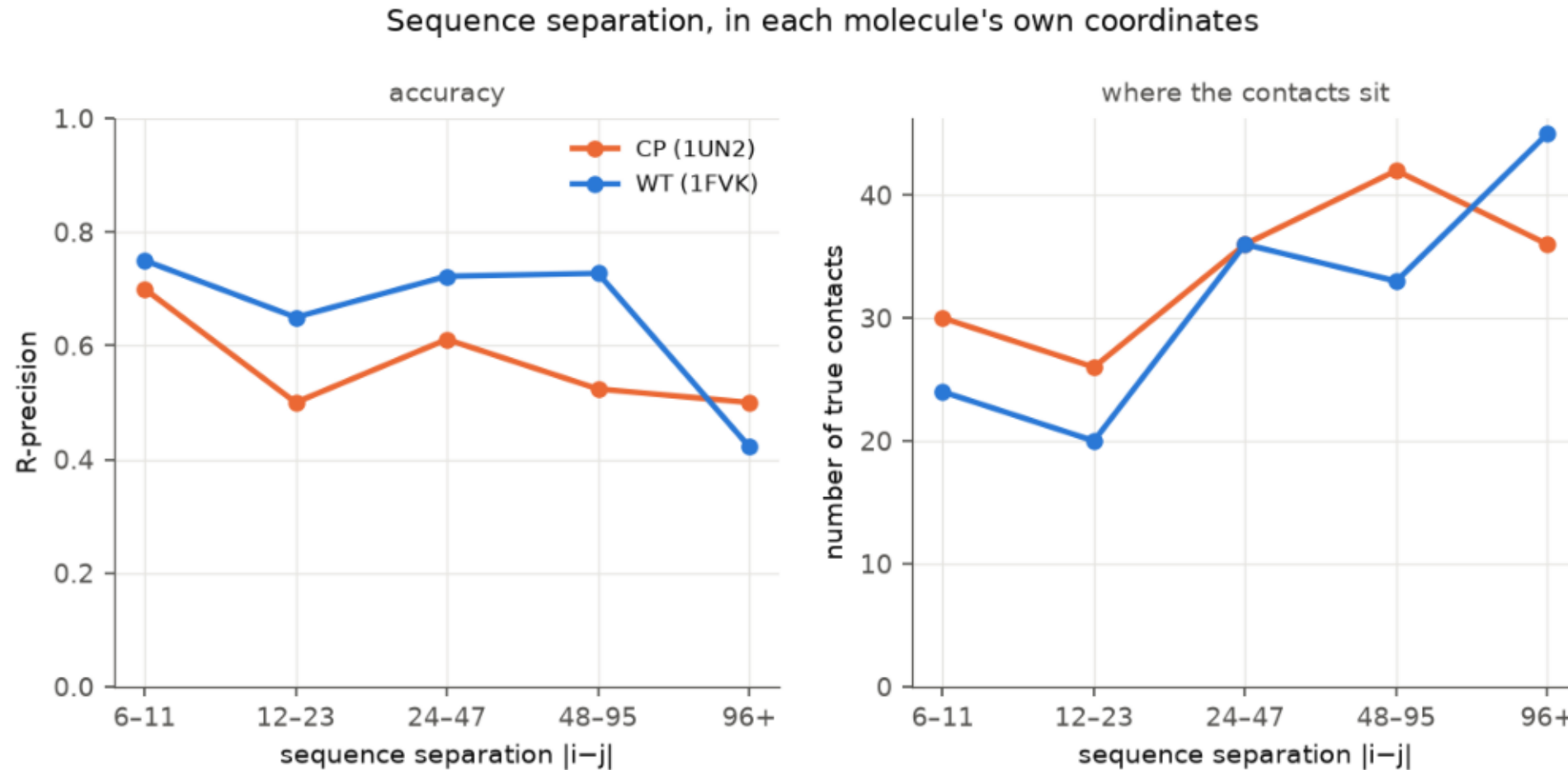
Same 3D contacts, same model — split by whether the permutation moved the pair



Error bars are the standard deviation over independent rollout seeds.

separation_profile.png

Accuracy by sequence separation (left) and where the true contacts sit on that axis (right), in each molecule's own coordinates.

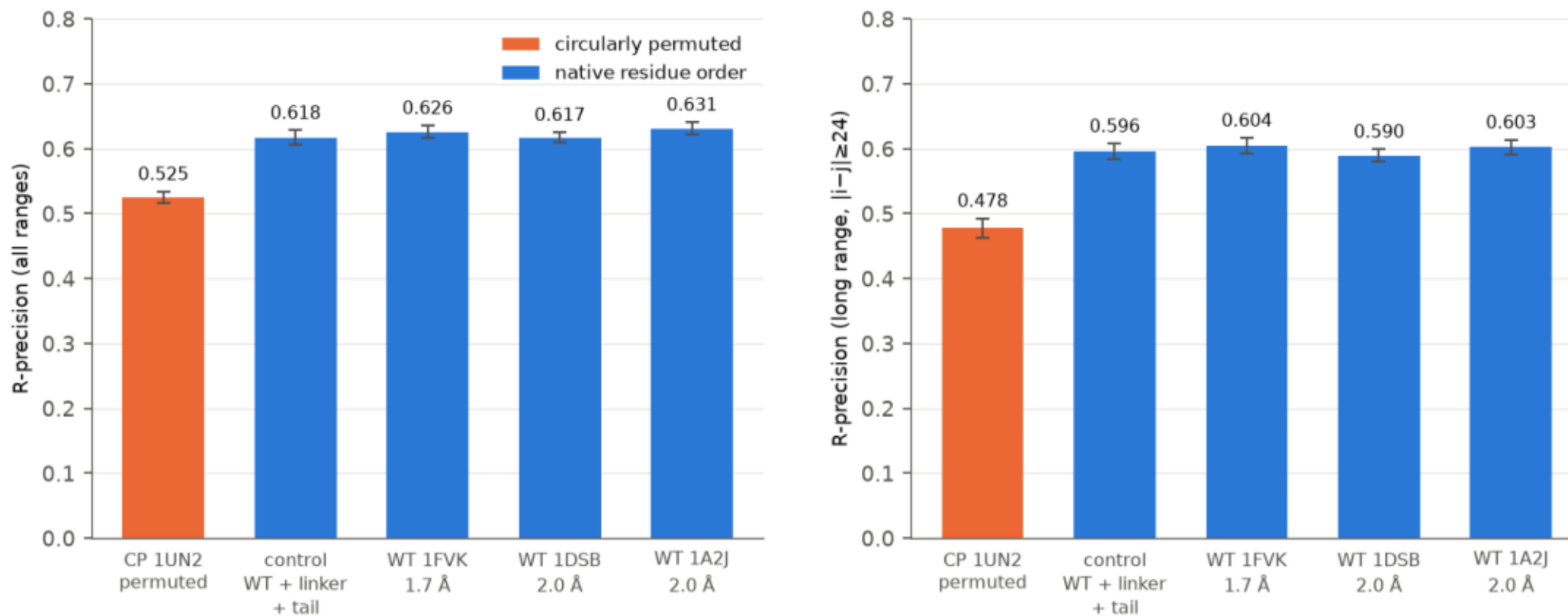


Same protein, same fold: the permutation moves ~47% of the contacts to a different separation (right), and those are the ones the model gets wrong (left).

summary.png

R-precision for the permutant, a length/composition control, and three wild-type crystals. The re-ordering costs ~ 0.09 ; the extra 8 residues cost nothing.

The re-ordering costs accuracy — the extra 8 residues do not



The control has the CP construct's exact length and non-native residues (GGGTG linker, LIK tail) in wild-type order. Error bars: sd over 10 rollout seeds.

The top-L predicted pairs, coloured by correctness, over the greyed-out truth. Precision 0.47 (CP) vs 0.57 (WT).

